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Polyphase magnetization in Lower Carboniferous rocks of S. Scotland: palaeomagnetic and tectonic considerations

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ABSTRACT

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Lower Carboniferous lavas and intrusives from the southern margin of the Midland Valley, Scotland, have yielded a principal magnetization, termed the B component, with D, $I = 192$, $+12$; $\alpha_{95} = 5.9°$. This magnetization is thought to have formed essentially during the original cooling, but evidence of low temperature mineralogical changes as well as the presence of dual-polarity remanence structures suggest that the magnetization processes continued beyond the deuteric stage, covering at least the time span of geomagnetic field inversions. An overprinted magnetization, the A1 component, of inferred Lower Tertiary age, has a mean direction of D, $I = 188$, -49; $\alpha_{95} = 3.5°$. This secondary magnetization is found principally in three sites, two from silica veins and one from a basaltic neck, located within the Southern Uplands Fault Zone at Dunbar. Microscopic evidence suggests that the A1 overprint was acquired thermochemically through Alpine tectonic re-activation of the fault.

1. Introduction

Both topographically and geologically South Scotland consists of two distinct regions: the Midland Valley and the Southern Uplands. The Midland Valley is bounded by two major crustal dislocations, the Highland Boundary Fault and the Southern Uplands Fault (see Fig. 1a). These tectonic lineaments are regarded as sinistral shear zones, at least with respect to their tectonic development in Caledonian times (e.g. Bluck, 1985; Hutton, 1987). The scale of the hypothesized strike–slip motions is unknown, but repeated events of diastrophism have turned the Midland Valley into a rift zone of complex geological history (e.g. Cameron and Stephenson, 1985).

The Lower Old Red Sandstone (ORS) is widely exposed in the northern Midland Valley and along its southern margin. Associated with the Lower ORS sediments there are intrusive and extrusive rocks, ranging in composition from basic to acidic. On the other hand, rocks of Middle ORS age are not recognized in the Midland Valley; the 'mid' Devonian Acadian orogeny apparently uplifted and deformed the basin prior to deposition of the Upper ORS. By the start of the Carboniferous Era the region had been worn down to low-lying, afforested coastal plains, similar to mangrove swamps of today (McAdam and Clarkson, 1986). Thus the Lower Carboniferous sedimentary succession, which is conformable with that of the Upper ORS, represents a cyclic fluvial–deltaic and shallow marine sequence consisting of coals, shales, mudstones and sandstones. Sedimentation was influenced by syn-depositional crustal unrest which gave rise to major volcanism, producing the

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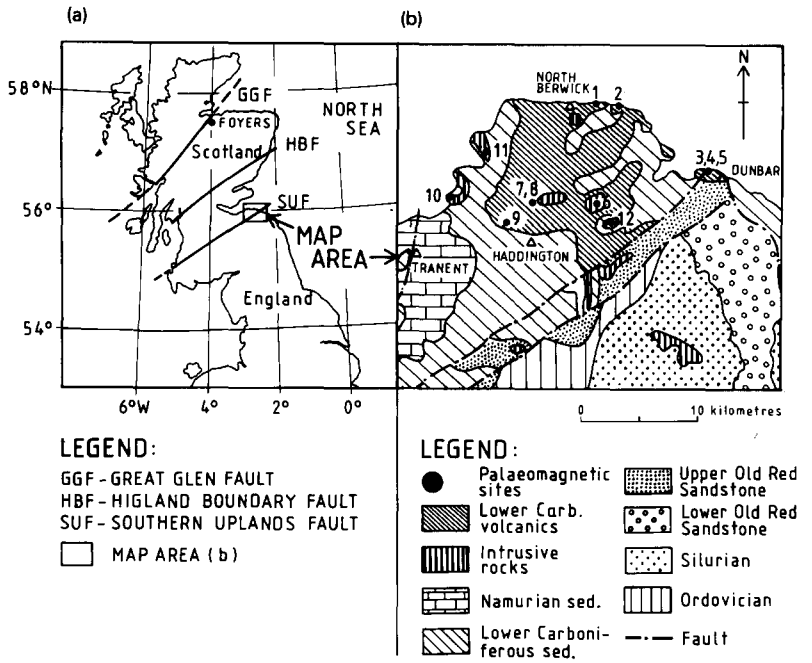


Fig. 1. Sketch maps showing (a) the geographic locations of the study area, relative to the main Scottish fault zones, and (b) the local geology. The geological map is a simplified version after McAdam and Clarkson (1986). The numbering refers to palaeomagnetic sampling sites.

characteristic NE–SW trending tectono-topographic grain of the region.

The Lower Carboniferous geomagnetic field relative to Britain has remained fairly unexplored, and therefore the wide variety of Midland Valley volcanics from this age interval are potentially of considerable palaeomagnetic interest. The pioneering study of these rocks was on the Kinghorn lavas, situated on the northern shore of the Firth of Forth (Wilson and Everitt, 1963). This early investigation suggested that the Lower Carboniferous field axis was practically identical to that for Old Red Sandstone times (cf. compilatory study by Storetvedt, 1967). A recent re-consideration of the Kinghorn lavas and other Midland Valley volcanics of the same age (Torsvik et al., 1989) in part confirmed the earlier findings, but some aspects of the new data indicate that the magnetization history may be more complex than hitherto envisaged (see discussion below). In the present paper we consider the magnetization structure of the Lothian volcanics, discussing palaeomagnetic aspects as well as some of the tectonic implications.

2. Rock sampling and laboratory details

The Lower Carboniferous igneous rocks considered here, collectively known as the Garleton Hills Volcanics, lie within the Calciferous Sandstone Measures of the Dunbar and Haddington districts. Recent geological overviews have been given by McAdams and Tulloch (1985) and Davies et al. (1986). The lower part of the volcanic succession is basaltic in composition, while the overlying extrusives constitute mostly thick (> 20 m) trachytic lavas. The majority of the intrusions are sills, but several volcanic plugs form prominent topographic features. Of the larger intrusive bodies the Traprain Law Laccolith has become a classic. Summary of radiometric ages (McAdam and Tulloch, 1985) suggests a Tournaisian–Visean origin of the main volcanic phase, in agreement

with the lithostratigraphic evidence. The principal tectonism of the Lower Carboniferous rocks is thought to date from the climax of the Hercynian orogeny (McAdam and Tulloch, 1985), causing an overall gentle folding along NE–SW axes.

Due to an extensive drift cover inland exposures are few and of poor quality, the main natural outcrops occurring in coastal sections. However, most natural exposures are affected by surface oxidation and weathering decomposition, so the majority of the palaeomagnetic samples were collected in working and disused quarries. Distribution of the palaeomagnetic sites is shown in Fig. lb, and further collection details are presented in Table 1.

In the laboratory, cylindrical specimens, with a diameter and height of 1.9 cm, were cut for palaeomagnetic measurements. About 50% of the samples were treated in the Bergen laboratory, employing a Schonsted TSM-1 (thermal) and a non-commercial two-axis alternating field (AF) demagnetizer, the remanence parameters after each demagnetization step being determined by Digico or Molspin magnetometers. The second half of the specimens was analysed in Potsdam where a two-axis alternating field demagnetizer and a JR-4 spinner magnetometer were used. For a minor series of samples AF demagnetization was topped by thermal treatment. On the whole, this latter procedure was unsuccessful in that an irregular (unstable) behaviour appeared at an early stage of thermal demagnetization.

The demagnetization results from the two laboratories proved to be closely comparable, and therefore the two specimen groups will not be identified separately. The overall remanence structure proved to be multicomponent, and for analytical purposes a combination of equal area or stereographic plots, orthogonal vector diagrams, and the vector subtraction technique have been used. Identification of magnetic minerals and their mode of occurrence has been by microscopic examination of polished sections.

3. Data acquisition

3.1. Demagnetization analysis

The palaeomagnetic success of the collected rocks is closely related to the degree of visible secondary alteration. On the whole, macroscopically fresh and hard material possesses the simplest and most systematic variation of measured remanence parameters vs. progressive demagnetization, providing practically all of the palaeomagnetic data obtained. However, only occasionally do these specimens possess a single characteristic

TABLE 1

Sampling details

Sites [a]	Sample numbers (NB)	Rock description
1	1–10	Taking Head Sill, North Berwick. Monchiquite. Rock is very fractured and partly weathered
2	11–20	Primrose Bank Sill. Purple basalts. Weathered
3 and 4	H21–27 [b]	Sampled in two 15-cm-thick veritcal veins cutting the reddened and decomposed columnar basalt at the Battery, Dunbar Harbour. The veins consist of hard red-coloured felsic material
5	28–32	Coastguard Station Neck. Purple basalt with extensive discolouration and haematitic veining
6	33–43	Pencraig Wood Trachyte Intrusion. Disused quarry Exposure affected by weathering alteration throughout
7 and 8	H44–50	Skid Hill Quarry. Trachyte lava. The two sites are about 80 m apart
9	H51–53	Bangley Quarry. Trachyte lava
10	54–63	Gosford Bay Sill. Hard and massive olivine dolerite exposed at tide level
11	64–68	Lower Gullane Head Sill. Massive and coarse teschenite in disused quarry
12	H69–73	Traprain Law Phonolie (laccolith)

[a] See Fig. 1 for locations.

[b] Samples denoted H are oriented hand samples. All other samples are oriented drill cores.

magnetization. The specimens frequently possess dual-polarity structures, demonstrated either by systematic directional trends during progressive demagnetization, or by trajectorial by-pass of the origin when the results are viewed in vector diagrams. Samples from sites that show clear evidence of secondary alteration, first of all our sites 1,2,5 and 6, gave in general too unsystematic (erratic) data to allow meaningful extraction/evaluation of their remanence build-up.

A very broad coverage of demagnetization data, outlining the remanence structure at specimen level, is depicted in Figs. 2–7. In total, the rocks have two characteristic axes of magnetization, a main B magnetization ($D_m = 192$, $I_m = +12$) and a subordinate A1 magnetization ($D_m = 188$, $I_m = -49$), that are clearly discriminated by direction. Both axes of magnetization have dual-polarity structure, but the reverse remanences predominate. Remanence notations (A1 and B) are according to terminology adopted for corresponding multicomponent magnetizations in Scottish Devonian rocks (e.g. Storetvedt et al., 1990).

Figures 2 and 3 present results from two relatively coarse-grained but fresh basaltic sills, sites 10 and 11 (Table 1). In the field these rocks were judged the most promising material of the entire collection. The experimental data from these two locations are in fact very clear-cut, defining unequivocally the reverse B magnetization. However, in most cases the trend of the vector trajectories by-pass the origin, signifying that the most resistant component of these two sites is a low intensity vector of normal polarity. In such cases, i.e. where the reverse magnetization is in dominance, the normal 'end-point' direction is precluded from experimental determination owing to increasing noise in the data at more advanced stages of demagnetization (to avoid cluttering of the diagrams this erratic stage is not depicted). On the other hand, our Traprain Law site (no. 12) has a preponderance of normal magnetization, and Fig. 4 shows the opposite situation of that for sites 10 and 11, i.e. a normal component is retrieved while the suggested reverse remanence remains experimentally unresolved.

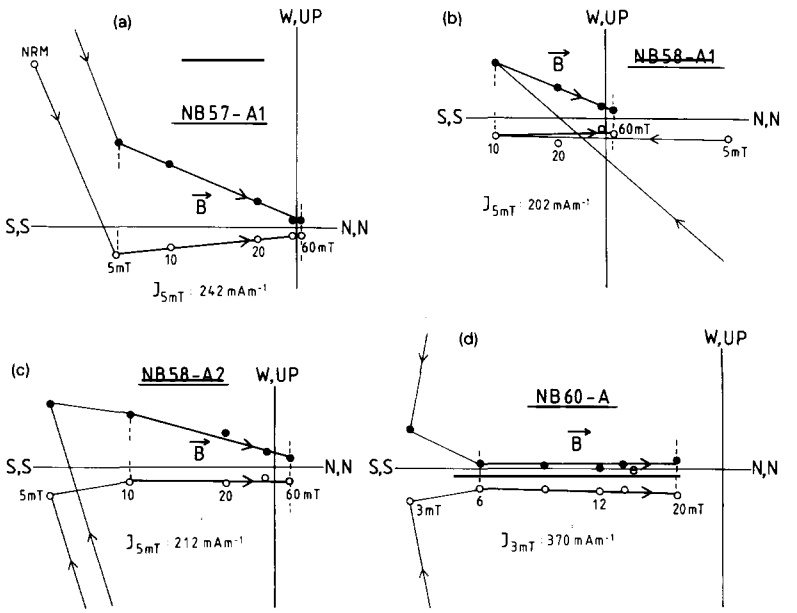


Fig. 2. Vectorial displays (orthogonal plots) of the predominant magnetization, the B component, of the Lower Carboniferous igneous rocks of East Lothian. Results are from site 10, Gosford Bay Sill. Solid (open) symbols are points in the horizontal (vertical) plane. Demagnetization ranges for characteristic remanence components are delineated by broken vertical lines. Magnetization intensities (J values) are given for indicated stage of demagnetization.

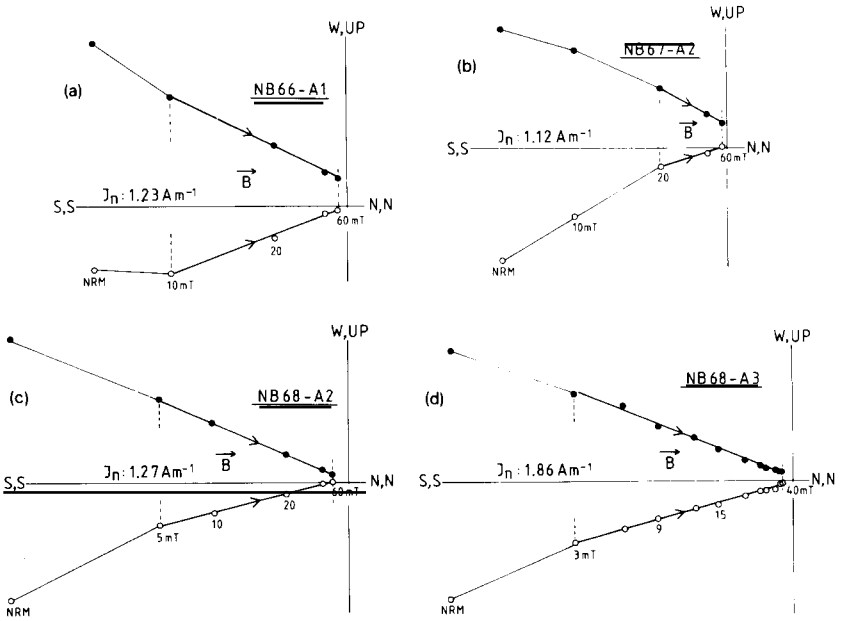


Fig. 3. Further orthogonal vector projections of specimens carrying the reversed B magnetization as principle remanence component. J_n denotes intensity of total remanent magnetization. Data from site 11, Lower Gullane Head Sill. Conventions are as for Fig. 2.

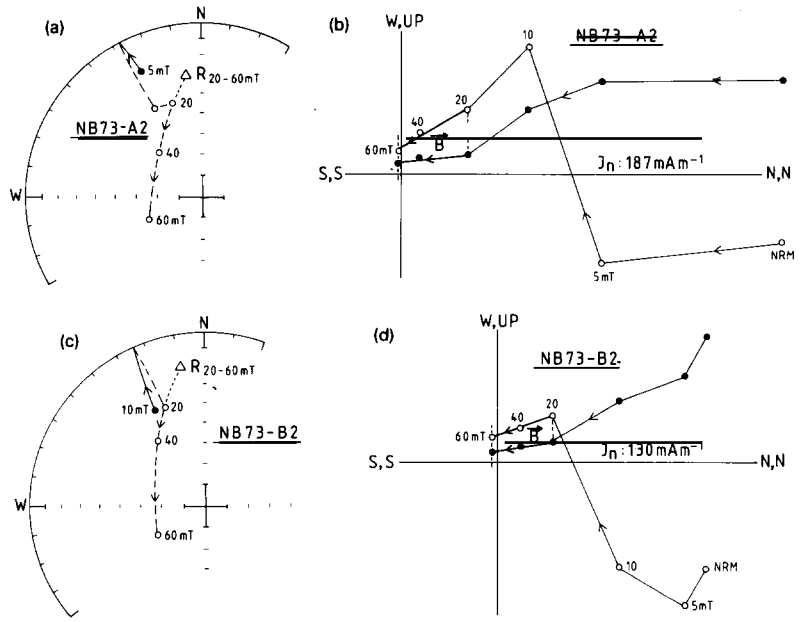


Fig. 4. Examples of the remanence structure of site 12, Traprain Law Phonolite, in which the characteristic magnetization has predominantly normal polarity. Diagrams (a) and (c) are equal area plots in which closed (open) symbols represent downward (upward) inclinations. Triangles (in (a) and (c)) denote directions of erased vectors for demagnetization range specified. Other conventions are as for Fig. 2.

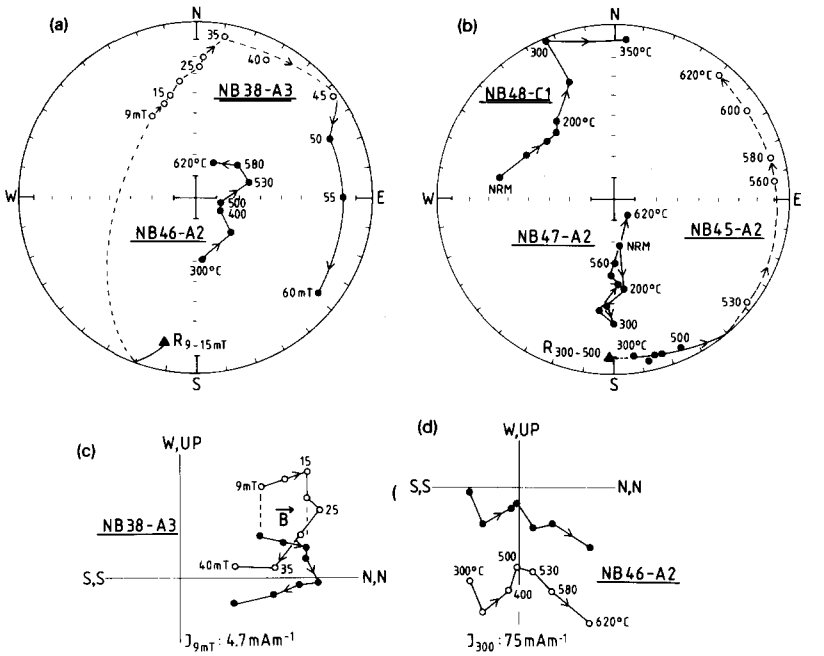


Fig. 5. Examples from sites 6–8 of dual-polarity interplay demonstrating variable component overlap. Co-existing normal and reverse magnetizations are commonly encountered in these Lower Carboniferous rocks, a feature that is also depicted in most examples of Figs. 2–4 by trajectorial by-pass of the origin. See text for general discussion of results. Other diagram conventions are as for Figs. 2 and 4.

Figure 5 exemplifies a larger group of specimens in which the normal and reverse components apparently are more strongly interleaved than in the examples discussed above, displaying only occasionally 'windows' in the demagnetization spectrum that allow extraction of one of the components (see 38-A3 and 45-A2). Specimen 38-A3, (a) and (c), illustrates the intertwined nature of the suggested antiparallel B components. Neglecting the fraction of the NRM with lowest stability a north trending remagnetization circle is traced out by successive demagnetization steps between 9 and 35 mT. A coercivity window in the 9–15 mT range allows the reverse B direction to be determined, but for higher demagnetization fields the dual-polarity interaction is sufficiently strong to prevent extraction of individual components. However, the remagnetization circle changes its orientation at around 35 mT treatment, above which the remanence vector performs a progressive southward movement. Although the 'endpoint' direction remains undecided, it appears reasonable to conclude that the reverse B component resides in grains of both low and high coercivity, while the normal B remanence, which apparently carries the substantial part of the fossil magnetization, covers a broader intermediate coercivity range. The intersection point of the two remagnetization circles (at 35 mT), which is nearly perfectly antiparallel to the direction of the vector removed in the 9–15 mT interval, may well correspond to the normal B magnetization. However, such results are too tentative for inclusion in a palaeomagnetic database.

A shallow inclined palaeomagnetic axis, constituting a dual-polarity remanence structure, can in principle produce stable or quasi-stable composite magnetizations in any spatial orientation. Also, conventional vector analysis may lead to erroneous conclusions. For example, specimen 46-A2 (Fig. 5a) performs part of a small circle motion in the 500–620°C range, being apparently in the course of reversing its northward trend, but the standard vector plots of Fig. 5d give reasonable

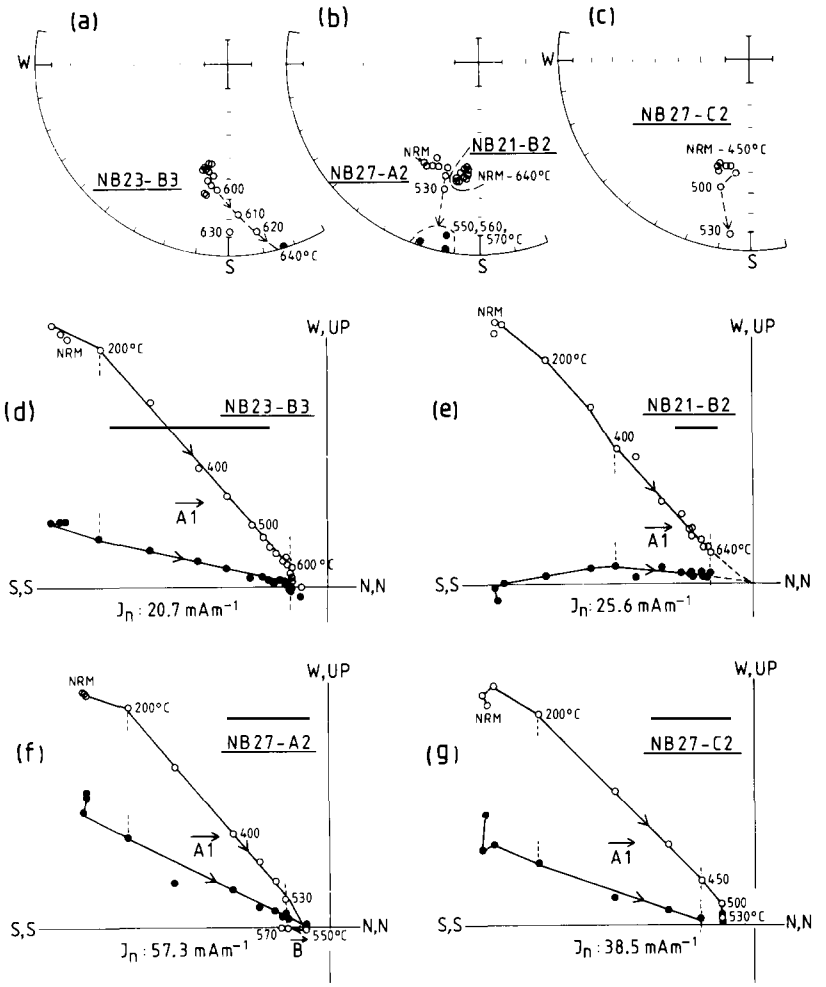


Fig. 6. Demagnetization data from sites 3 (specimens 21-B2 and 23-Bc) and 4 (specimens 27A2 and C2) depicting a second palaeomagnetic component, the A1 magnetization, in addition to an inferred weak high blocking temperature B remanence (see a–c and f). Diagram conventions are as for Figs. 2–4.

'linear' trends that in an uncritical approach may incorrectly be given palaeomagnetic relevance. If not properly understood such stray 'components' may cause considerable confusion in palaeomagnetic studies. In fact, interleaved normal and reverse magnetization spectra frequently display a 'forward and backward' type of directional behaviour, from which none of the inherent palaeomagnetic components can be retrieved.

A second axis of magnetization, A1, has been found in four of the sites, 3, 4, 5 and 7, the principal locations being nos. 3 and 4 (i.e. the two felsic veins collected at Dunbar Harbour). Figure 6 depicts demagnetization data from sites 3 and 4 in which haematite, according to thermal demagnetization, is the predominant remanence carrier. The A1 magnetization, defined by intermediate up inclinations and nearly due south declination, is occasionally the only component detected (see 21-B2, Figs. 6b and 6e) but in general thermal demagnetization above about 550°C provides shallower inclination, the remanence vector moving towards/into the reverse B-axis region. The final end-point directions are associated with low remanence intensity and therefore they are difficult to establish, but specimen 27-A2 (Figs. 6b and 6f)

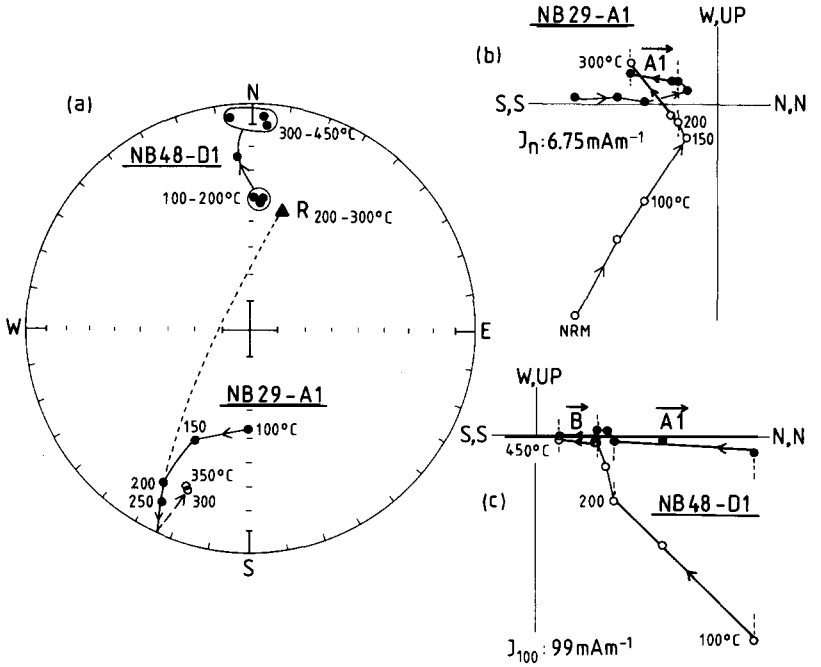


Fig. 7. Further examples of co-existing B and A1 magnetizations (in addition to those of Fig. 6) from sites 5 (specimen 29-A1) and 8 (specimen 48-D1). Diagram conventions are as for Figs. 2, 4 and 5.

might have arrived at its terminal high temperature end-point.

Further examples of magnetization A1 are shown in Fig. 7. Specimen 48-dl (a and c) apparently contains the normal A1 and B components in co-existence, while 29-A1 (a and b) seems to carry dual-polarity A1 magnetization. For the latter specimen, however, an end-point direction (reversed) is not properly confirmed.

In conclusion, the present study has uncovered two distinctly different magnetization axes, A1 and B, both with dual-polarity structure. The numerical data for these remanence populations are listed in Table 2.

TABLE 2

Overall data for magnetization axes of Lower Carboniferous rocks of East Lothian in conjunction with 'corresponding' magnetizations of the Foyers Old Red Sandstone, Inverness-shire (Storetvedt et al., 1990). Previous data from Lower Carboniferous volcanics of the Eastern Midland Valley (Wilson and Everitt, 1963; Torsvik et al., 1989) are also included

Component	Suggested	N, n	D_m	I_m	α_{95}	Pole	Reference
A1	M/T	15	188	−49	3.5	162E, 63N	This paper
A1	M/T	18	018	48	4.3	144E, 50N	Storetvedt et al., 1990
B	C_1	33	192	12	5.9	164E, 27N	This paper
B	D_{m-u}	81	185	8	2.7	170E, 29N	Storetvedt et al., 1990
'B'	C_1	17 (n)	196	28.5	4.4		Wilson and Everitt, 1963
B	C_1	13 (n)	205	31	7.0	152E, 14N	Torsvik et al., 1989

M, T, C and D denote Mesozoic, Tertiary, Carboniferous and Devonian respectively. Other symbols are; l, lower; m, middle; u, upper; N, number of unit vectors (specimens); D_m, mean declination, I_m, mean inclination; α_{95}, radius in the 95% circle of confidence; N, number of specimens; n, number of lava flows or sites.

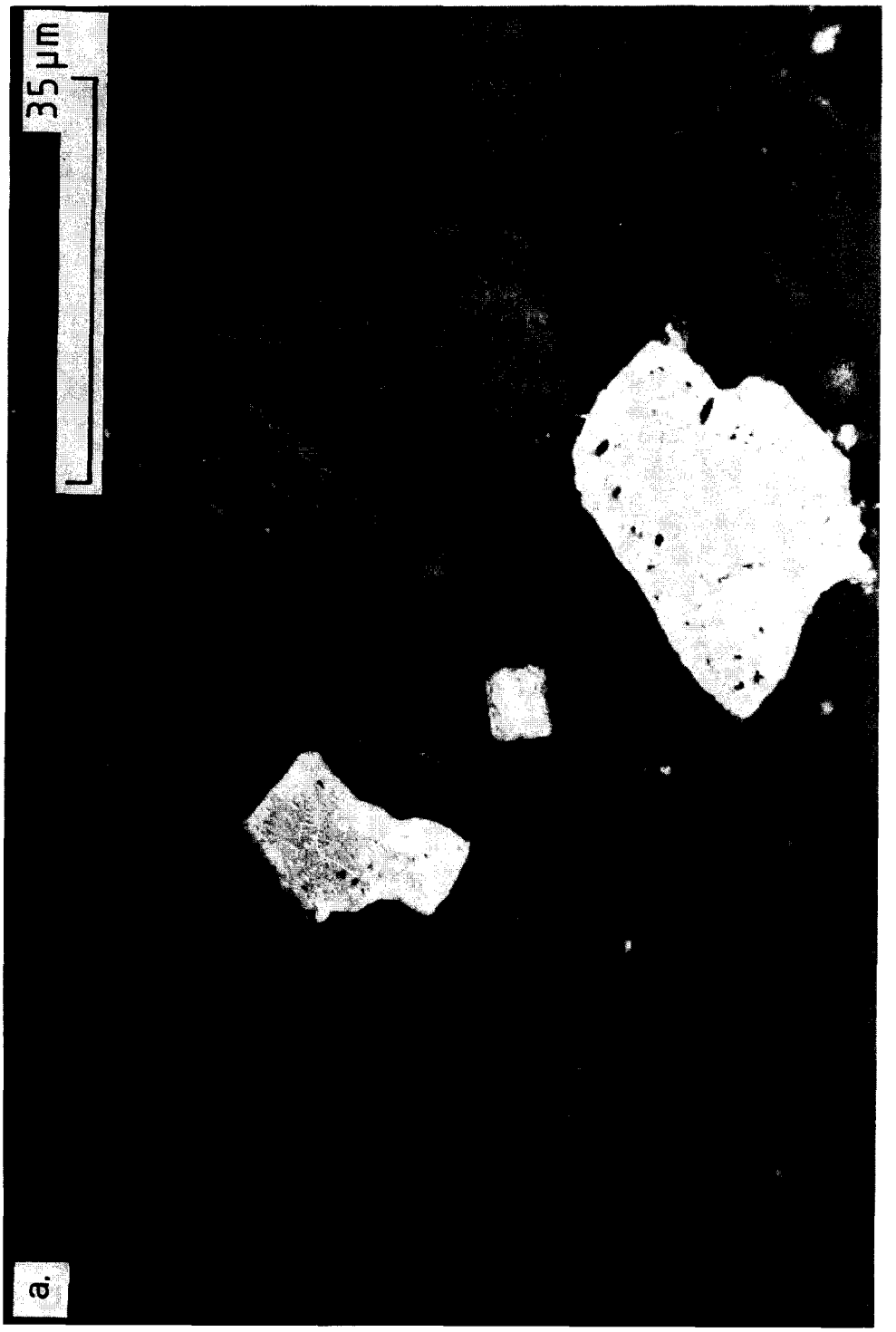


Fig. 8. Photomicrographs of opaque grains in polished sections: (a) idiomorphic magnetite grains from site 7 showing the development of martitization, haematite (white) now occupying nearly 50% of individual magnetite grains;(b) large magnetite grain from site 11 in a fairly advanced state of secondary alteration.



Fig. 8 (continued).

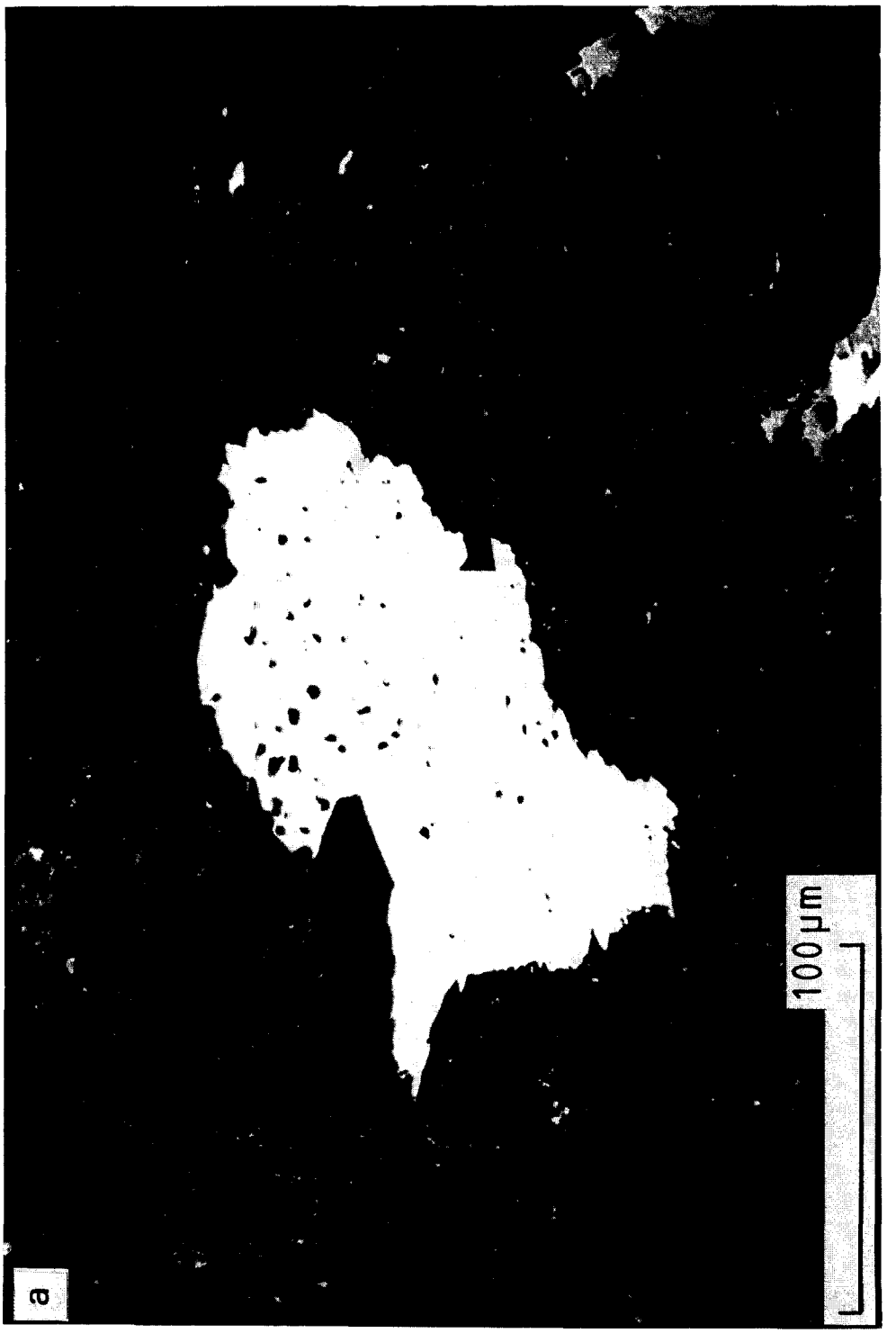


Fig. 9. Further photomicrographs showing: (a) polycrystalline haematite embedded in a red haematitic silicious matrix of site 3; (b) example of tectonized ('brecciated') groundmass of site 3 associated with a partly glassy (dark) matrix. The dark amorphous matrix is interpreted as evidence of stress-induced high temperature remelting. The ring of small secondary haematite grains (white) should be noted, which have evolved around the remelted matrix.

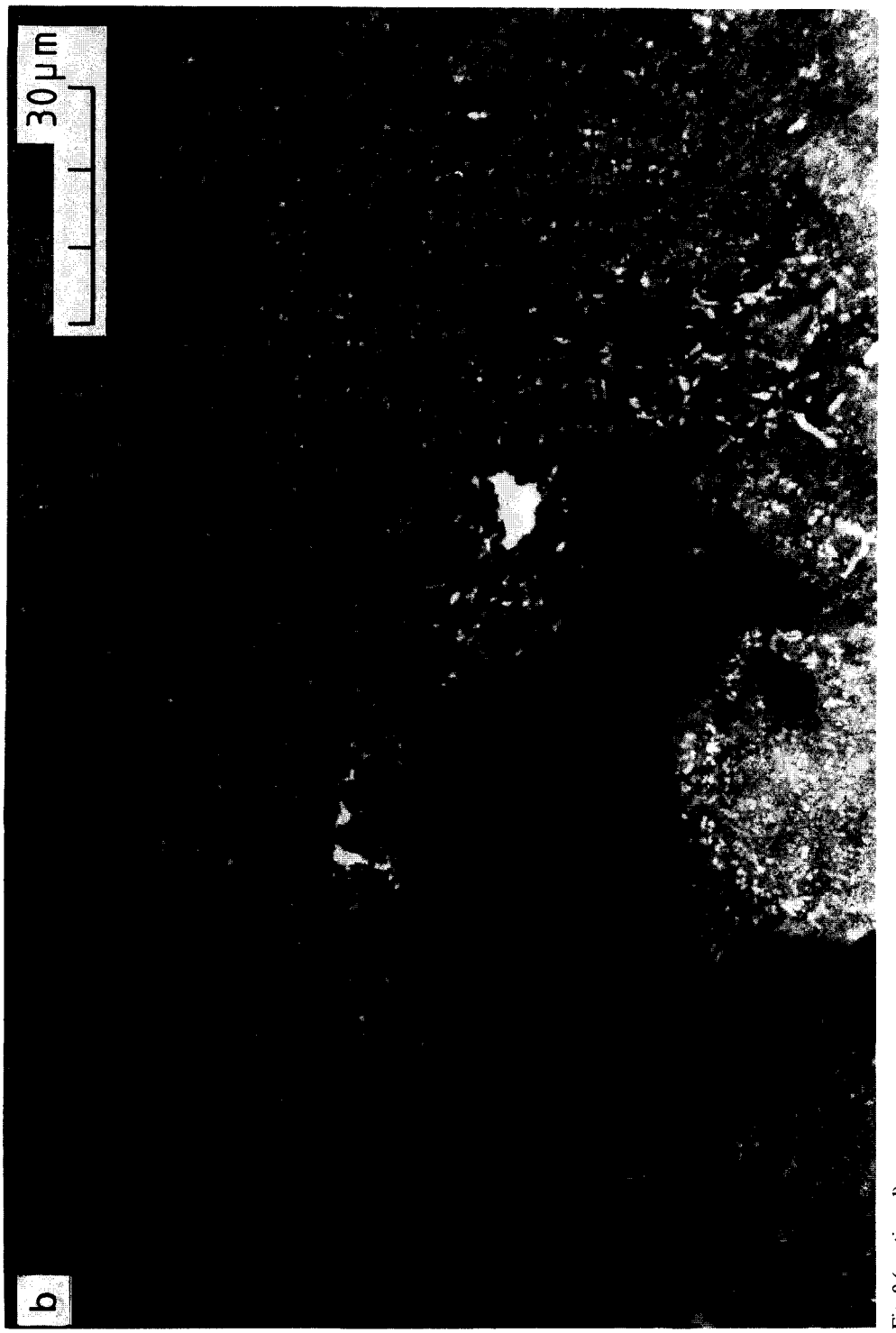


Fig. 9 (continued).

3.2. *Magnetomineralogical properties*

Microscopic examination of polished sections show them to be quite variable in terms of the amount, grain size and degree of alteration of their opaque minerals. Individual oxide grains display a wide range in the degree of oxidation. Martitisation, exemplified by Fig. 8a, is commonly observed, and smaller magnetite grains in particular have sometimes suffered complete conversion to haematite. In addition to martite, which may form at both high and low temperature (Ramdohr, 1969), more specific low temperature oxidation of the primary magnetite, characterized by contraction cracks and irregular mineral alteration (Fig. 8b), is frequently seen. However, secondary alteration is noticeably less developed in the Gosford Bay and Lower Gullane Head sills (sites 10 and 11) which are the two rocks in the collection that carry the largest proportion of homogeneous, idiomorphic magnetite grains. The silica veins, represented by sites 3 and 4, are dominated by haematite. Finely dispersed haematite dust gives the veins a deep red colour, but scattered throughout the fine-grained silicious–haematitic matrix haematite particles of variable grain size occur (see Fig. 9a). An observation of particular interest is shown by Fig. 9b in which a tectonically disrupted part of the groundmass occurs in association with a partly glassy matrix. The dark amorphous matrix is regarded as the product of stress-related high temperature partial melting, giving rise to the growth of a second generation of small haematite particles in the adjacent non-melted groundmass. With regard to magnetic mineralogy, sites 3 and 4 are markedly different from the volcanics as can be demonstrated by, for example, the hysteresis loops of Fig. 10. Site 5 (the purple basalt of the Coastguard Station Neck) is the only volcanic site in this study for which haematite is the dominating iron oxide. As can also be seen from Fig. 1, sites 3–5 have one common feature, distinct from the other sites, by being located within the Southern Uplands Fault Zone.

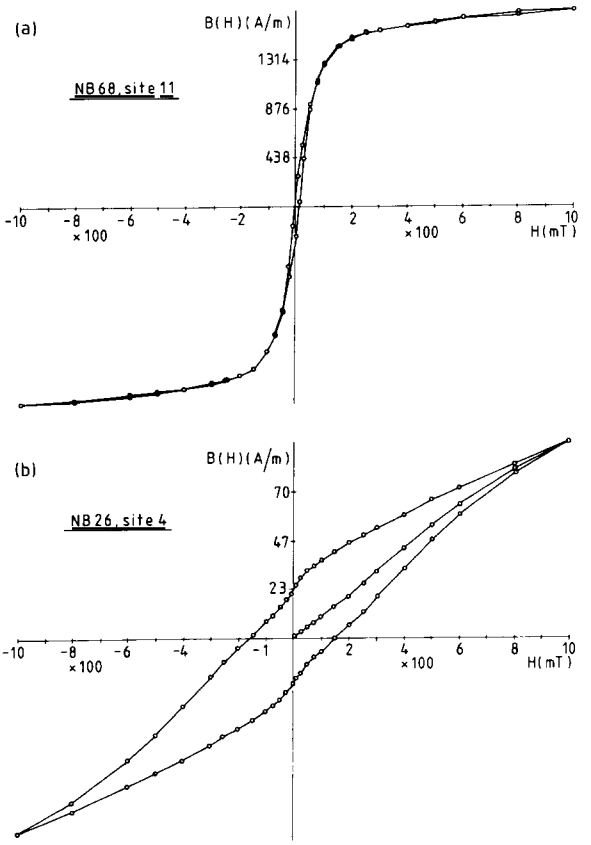


Fig. 10. Hysteresis loops demonstrating extremes in iron oxide content: a, sample from site 11 that has a preponderance of 'magnetite'; b, sample from site 4 that is entirely dominated by haematite.

4. Interpretation

4.1. *Palaeomagnetic aspects*

The shallow B component, which contributes the predominant fossil magnetization of the present study (Fig. 11a) has in situ mean direction $D/I = 192/12$, $\alpha_{95} = 5.9°$. The B remanence has dual-polarity structure, but apart from site 12 which is governed by the normal polarity component, the reverse magnetization predominates. To judge from the overall fresh state of the iron oxide mineralogy of sites 10 and 11, along with their very clear-cut B component structure, it appears reasonable to conclude that the B magnetization principally dates from the time the rocks cooled. However, the presence of low temperature alteration of the original magnetic mineralogy, occur-

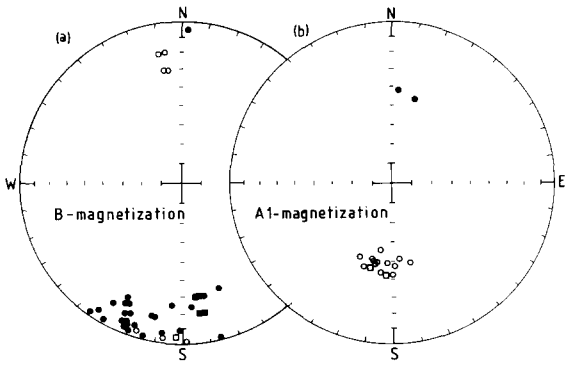


Fig. 11. Equal area plots of characteristic remanence directions at specimen level. Diagram (a) depicts the distribution of the principal B magnetization while diagram (b) shows the overprinted A1 component. Squares are inverted normal polarity results. Overall palaeomagnetic data for the two populations are listed in Table 2. Conventions are as for Fig. 4.

ring in varying degree in all our igneous sites, along with the dual-polarity build-up of the B magnetization, suggest that acquisition of low temperature (late-post cooling) chemical remanence operated over time ranges spanning at least geomagnetic field inversions. Furthermore, it is necessary to point out that our B population has a certain 'undesirable' spread in declination (see Fig. 11a) that is most likely the effect of either incomplete component resolution or local tectonic rotation (see discussion below).

It is generally assumed that the principal tectonic structures of the Lower Carboniferous rocks of East Lothian date from the climax of the Hercynian Orogeny in late Carboniferous–early Permian time. However, except in the proximity of faults the stratal disturbance is very gentle. Thus, in the coastal area of the Garleton Hills volcanics, where the regional tilt is westerly, the tectonic dip is only around 5°. On present knowledge, therefore, the Hercynian deformation would only cause insignificant changes in pre-tectonic B magnetization. Consequently, we regard the overall in situ B direction (see Table 2) to be in close approximation to the average Lower Carboniferous geomagnetic field direction relative to South Scotland.

The characteristic palaeomagnetic axis arrived at here is about 15° shallower than that previously reported from corresponding rocks at Kinghorn, on the opposite shores of the Firth of Forth (see Wilson and Everitt, 1963; Torsvik et al., 1989). In Fig. 12 we have reproduced some data from Torsvik et al. (1989) that may possibly throw light on this discrepancy. Figure 12a shows the distribution of their site mean directions for the Kinghorn lavas in conjunction with the remagnetization circle through the overall palaeomagnetic axis established by the present study, and Fig. 12b delineates the same association for a 'two-component' lava in the Western Midland Valley. From the total spread of B directions (see Fig. 11a) a wide variety of remagnetization trends might be envisaged, but it is interesting to note how well many of the Torsvik et al. mean directions fall along our 'average' remagnetization circle (i.e. the great circle through the average dual polarity remanence axis). This suggests to us that at least some of the Torsvik et al. data (Fig. 12a or 12b) may suffer from incomplete experimental resolution. Torsvik et al. dismissed the results of their sites 27 and 30 without further consideration. However, we believe that one of the ground rules for assessing data in all empirical sciences should be that a holistic approach in interpretation be followed: without reasonably understanding the total database as a complete system one cannot in an objective way apprehend the physical meaning of selected results. An interesting analytical aspect is presented by specimen L 298-A in which the Torsvik et al. interpretation (Fig. 12c) is reassessed by us (Fig. 12d). Our qualitative re-analysis makes better use of the observations below 500°C, leading to reduced scatter along suggested vector lines. The important difference in the new interpretation is that a very shallow down direction with due south declination, consistent with our B magnetization, is readily apparent. The indicated high temperature component ($T > 530$°C) of Torsvik et al., which they consider as their primary (B) magnetization, is not seen in the data of the present study. Specimen L 298-A exhibits systematic directional variation at least up to 530°C, and because there is every reason to assume that the remanence is dual-polarity (as in the present study), artificial end-point magnetization with 'intermediate' inclinations can readily form.

The A1 magnetization, with $D/I = 188, -49$,

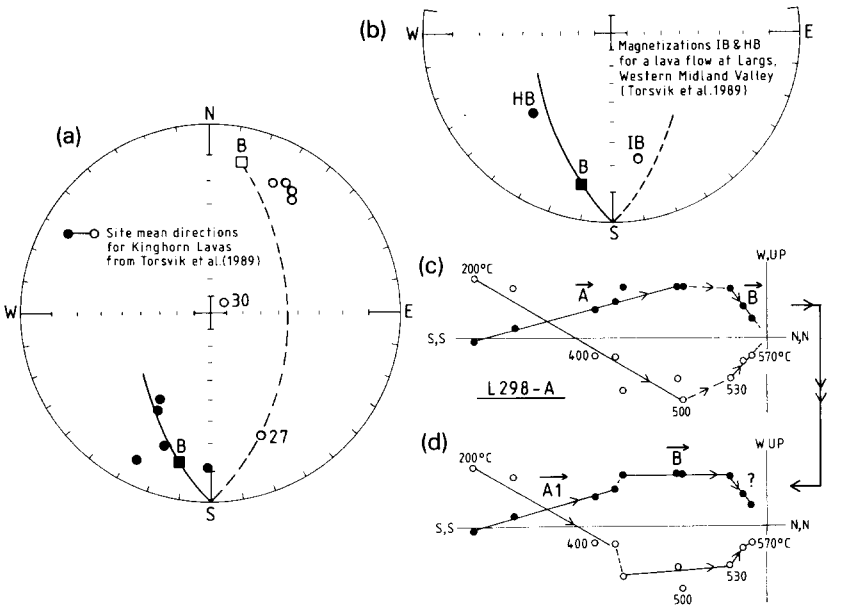


Fig. 12. Results of Lower Carboniferous lavas from the Midland Valley, according to Torsvik et al. (1989). Diagrams (a) and (b) (stereoplots) display site mean directions, the distribution of which is compared with the 'average' remagnetization circle (see text) through the dual-polarity B-axis established by the present study. Diagram (c) gives orthogonal vector diagrams of a thermally demagnetized lava specimen, while (d) suggests a component interpretation (of the same specimen) different from that of the original authors (c) (see text for more detailed discussion).

$\alpha_{95} = 3.5°$, has been found in four sites (nos. 3, 4, 5 and 7) of which the two vein sites (nos. 3 and 4) are the principal ones. Recent consideration of the Meso-Caenozoic palaeomagnetic database for Europe (Storetvedt, 1990) favours a Lower Tertiary age for the A1 magnetization. The fact that the A1 component co-exists with the B remanence is further evidence of the secondary nature of this magnetization. In previous studies of Scottish rocks magnetizations equivalent to the A1 component have been uncovered from the loosely consolidated John O'Groats Sandstone (Storetvedt and Carmichael, 1979), the Duncansby Neck (Storetvedt et al., 1978) and the Foyers Sandstone (Storetvedt et al., 1990). Comparison of the present A1 and B magnetizations with those of the Foyers Sandstone is given in Table 2.

4.2. Tectonic consequences

The veins that house our sites 3 and 4 have a strike direction of about 045°. This NE–SW orientation which corresponds to a main set of joints at Dunbar, is consistent with the direction of the major fault system that cuts through Scotland and Northern England. The A1 magnetization and other older overprints in rocks adjacent to the Great Glen Fault have been associated with repeated tectonic reactivation of the fault (Storetvedt and Meland, 1985; Storetvedt, 1987; Storetvedt et al., 1990). The observation of a strong A1 magnetization in sites 3–5, located within the Southern Uplands Fault (SUF), provides further manifestation of Alpine age fault rejuvenation in Northern Britain. Both the Great Glen Fault, the Highland Boundary Fault, and other major dislocations across Britain have repeatedly been regarded as important strike–slip fractures (e.g. Storetvedt, 1974, 1987; Bluck, 1984; Hutton, 1987), and we believe that our present observation of partial crushing and re-melting at microscopic scales in sites 3 and 4 bear on an Alpine age shearing event along the SUF. It may also be pertinent to this discussion that the magnetization

directions from our Traprain Law site, i.e. the four normal up directions of Fig. 11a, show a certain 'axial' misalignment (in the counterclockwise sense) compared with the bulk of reversed directions from other sites (Fig. 11a). This 'discrepancy' may be accounted for by a minor tectonic rotation of Traprain Law, an explanation that is reasonable taking into account the close proximity (about 3 km) of the SUF (cf. Fig. 1). However, a much more comprehensive palaeomagnetic study of Traprain Law is needed to clarify this question.

Hercynian magnetic overprinting (A2-components, cf. Storetvedt et al., 1990) along the Great Glen Fault has repeatedly been reported during the past two decades, and exposures of Permo-Carboniferous intrusives can be found in many parts of Scotland, including the Midland Valley. However, Hercynian overprints have not been uncovered by the present study.

As can be seen from Table 2 the East Lothian B magnetization is not significantly different from that of the Foyers Old Red Sandstone of Lower Middle Devonian age. This is not taken as evidence of a Lower Carboniferous age for the Foyers B component but rather as an indication of average relative geomagnetic field constancy over a time span covering at least most of the Devonian and the Lower Carboniferous. The many different rock types in the Northern Highlands that have the flat lying B magnetization as their characteristic magnetization (see Storetvedt, 1987), render the remagnetization alternative very unlikely.

The fact that the B magnetizations across the Highland Boundary Fault are indistinguishable does not rule out significant transcurrent faulting along this major fracture zone in mid-late Palaeozoic time. Both the Southern Uplands Fault and the Highland Boundary Fault define fairly linear tectonic lines across Scotland, and therefore strike-slip faulting of the order of a few hundred kilometres would not show up by palaeomagnetic discordances astride these fracture zones. In fact, it is only the curved pattern of the Great Glen Fault that seems to have caused declination discordances across this dislocation and which can be used to unveil its displacement history (see Storetvedt, 1987; Storetvedt and Otterå, 1988).

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